

Macroeconomic and Financial Indicators: Cross-Country Exploratory Analysis

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Introduction

Data source and collection

The data come from the World Bank **World Development Indicators (WDI)** database (World Bank, 2026), which compiles national accounts, survey, and administrative records submitted by member countries to the World Bank, IMF, ILO, and OECD. Indicator codes were checked against the WDI R package (full details in `FindData.Rmd`). Rather than pulling from the live API, this report uses a fixed snapshot (`Dataset.csv`) downloaded from World Bank DataBank on 24 February 2026—covering 1960–2025—so all results can be reproduced offline.

Unit of observation

Each row is a **country–year pair**. After excluding World Bank regional and income-group composites, the full grid contains 14,322 country–year slots across 217 economies over 1960–2025; of these, 12,113 have at least one non-missing indicator value. Missingness rates reported below use the full 14,322 grid as the denominator, not the 12,113 active-observation count. The sample covers UN member states and dependent territories that maintain separate national accounts (e.g., Hong Kong SAR, Virgin Islands U.S.), so “economy” is the more accurate label throughout.

Variables, natural limits, and missing-value coding

All eight indicators are continuous numeric variables; the WDI package codes missing entries as NA rather than sentinel values like `–99`. Since no variable is categorical, checks for valid category membership and full category coverage are not applicable here. No cross-variable logical constraints apply either—each indicator is defined independently. Table 1 lists each variable alongside its unit and the natural limits any plausible value must satisfy.

Table 1: Variable definitions, units, and natural limits.

Variable	Description	Unit	Natural Limits
Credit/GDP	Domestic credit to the private sector	% of GDP	≥ 0 (no strict upper bound)
GDP Growth	Annual growth rate of real GDP	%	No strict bounds; $ x > 50$ is suspicious
GDP per Capita	Gross domestic product divided by population	Current US\$	> 0
Investment/GDP	Gross capital formation (fixed investment + inventories)	% of GDP	No strict lower bound under SNA 2008; extreme negatives warrant scrutiny
Inflation	Annual change in consumer price index	%	No strict bounds; $x > 500$ suggests hyperinflation
Market Cap/GDP	Market capitalisation of listed domestic companies	% of GDP	≥ 0 (no strict upper bound)
Trade/GDP	Sum of exports and imports of goods and services	% of GDP	≥ 0 (can exceed 100 for entrepôt economies)
Unemployment	Share of the labour force without work (ILO estimate)	%	$0 \leq x \leq 100$

Data Quality Checks

Missing data

Market Cap/GDP leads on missingness (81.2%)—stock-exchange data do not exist for countries without formal capital markets—and pre-1970 coverage is thin across most variables. The gaps are correlated: 52.8% of observations missing Market Cap/GDP are also missing Credit/GDP; 63.2% missing Credit/GDP are also missing Investment/GDP. Financial variables form a coherent missing-data block and should be imputed or filtered as a group.

Natural-limit violations and outliers

Unemployment and GDP per Capita respect their natural limits throughout. The minimum GDP per Capita (US\$11.8, Myanmar 1966) is an exchange-rate artefact; the maximum (US\$288,001, Monaco 2024) is plausible for a very small, very wealthy economy.

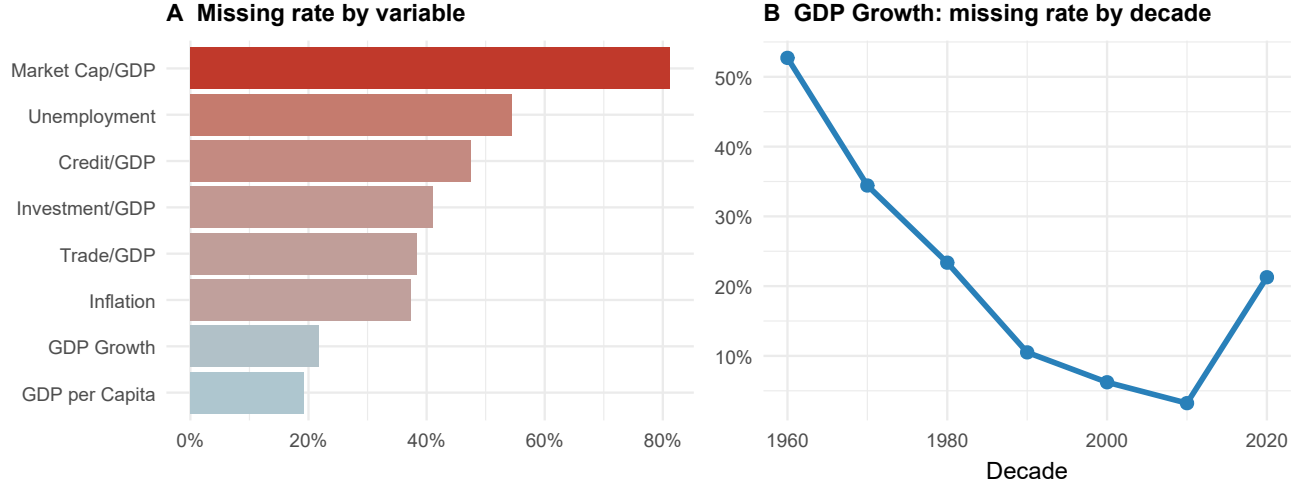


Figure 1: Left: percentage of country–year observations missing for each indicator. Right: share of economies with missing GDP Growth values, grouped by decade—illustrating that data coverage has improved markedly since the 1970s. Both panels are restricted to economy-level observations (aggregates excluded).

Negative Investment/GDP: 11 observations show negative gross capital formation (min -15.7%, Djibouti 2021). SNA 2008 permits this during large inventory drawdowns; values are retained but warrant scrutiny in investment-focused models.

Extreme GDP Growth: 22 observations exceed $|x| > 50\%$ (max 150%, Equatorial Guinea 1997). These fit post-conflict or oil-boom patterns—not recording errors—but carry high leverage in any regression.

Extreme Trade/GDP: Maximum 863% (Virgin Islands (U.S.), 2008). Values above 400% are plausible for entrepôt economies where re-exports far exceed domestic output, making the GDP denominator misleadingly small.

Extreme Credit/GDP: Maximum 301% (Iceland, 2006)—plausible for highly financialised economies (Hong Kong SAR, Cyprus) but a high-leverage point in credit regressions.

Extreme Market Cap/GDP: Ratios above 100% arise when multinationals list locally but operate abroad (e.g., mainland Chinese firms on the Hong Kong exchange); the inflated ratio is a denominator mismatch, not bad data.

Lower-tail zeros: The 0.0% minima shown for Trade/GDP, Credit/GDP, and Market Cap/GDP in Table 3 are display-rounding artefacts—the one-decimal-place format rounds small positive values down to 0.0. No observation is exactly zero for any of these variables (0 exact-zero rows for Trade/GDP; 0 for Credit/GDP; 0 for Market Cap/GDP), so no miscoding correction is needed.

Hyperinflation: 6 country–year observations record inflation above 500%.

Table 2: Country–years with inflation exceeding 500%.

Country	Year	Inflation (%)
Congo, Dem. Rep.	1994	23773.1
Bolivia	1985	11749.6
Peru	1990	7481.7
Ukraine	1993	4734.9
Angola	1996	4145.1
Congo, Dem. Rep.	1992	4129.2

Each entry in Table 2 is arithmetically valid and corresponds to a known currency crisis—the Congo, Dem. Rep. episode in 1994 being the most severe. These observations stay in the dataset, but log-transformation or winsorisation is strongly advisable before fitting any regression.

Data Summaries

Summary statistics

Table 3: Summary statistics for all eight indicators (individual country–years).

Variable	N	Min	Q1	Median	Mean	Q3	Max	NA (%)
Credit/GDP	7523	0.0	12.9	26.3	39.5	51.9	301.0	47.5
GDP Growth	11214	-64.0	1.2	3.8	3.6	6.3	150.0	21.7
GDP per Capita	11569	11.8	613.0	2,195.4	9,658.0	9,404.5	288,001.4	19.2
Inflation	8968	-17.6	2.1	4.8	22.8	10.1	23,773.1	37.4
Investment/GDP	8441	-15.7	18.0	22.7	23.3	27.6	76.8	41.1
Market Cap/GDP	2695	0.0	16.2	34.6	60.3	71.5	1,777.2	81.2
Trade/GDP	8819	0.0	44.6	66.7	78.2	97.0	863.2	38.4
Unemployment	6531	0.1	3.6	6.3	8.1	10.9	38.8	54.4

Distributions

Log-transformation is necessary for two variables (Figure 2). GDP per Capita spans roughly three orders of magnitude, and Inflation’s distribution still shows a pronounced right tail even on a log scale—a direct footprint

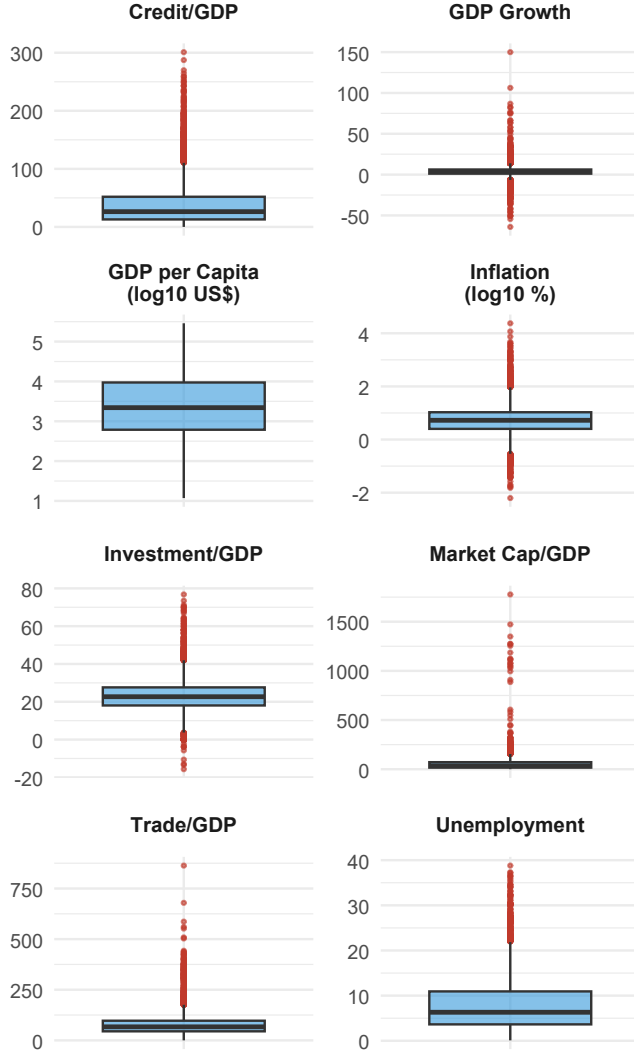


Figure 2: Distribution of each indicator across all country-year observations. GDP per Capita and Inflation are displayed on a log10 scale (positive values only) to handle extreme right skew; negative inflation observations are excluded from those panels only. Boxplot whiskers extend to 1.5 IQR; individual points beyond are drawn in red.

of the hyperinflationary episodes in Table 2. The other six variables sit more symmetrically around their medians, though none are entirely free of outliers.

Time-series trends by income group

Three patterns emerge (Figure 3). Credit/GDP has risen steadily in high-income economies since 2000 while stagnating in low-income ones—the gap keeps widening. Trade/GDP grew through the 1990s–2000s, stalled post-2008, and only partially recovered, consistent with Constantinescu, Mattoo & Ruta (2020). GDP growth synchronises sharply at both crisis windows; the 2020 contraction is the deepest in the sample.

Bivariate relationships

Figure 4 shows Credit/GDP and GDP per Capita as the tightest pair ($\rho = 0.74$), with Market Cap/GDP and Credit/GDP close behind ($\rho = 0.68$)—the familiar financial-depth cluster (King & Levine, 1993; Levine, 2005). These estimates are from a non-random ~19% subsample biased toward higher-income economies ($N \sim 2,700$ vs. $N \sim 11,000$ for non-financial pairs), so they may not generalise and some blank cells may reflect power limitations rather than true absence of association. Inflation and GDP Growth show essentially no rank correlation ($\rho = 0.03$), suggesting that the classical negative inflation–growth link operates primarily through extreme episodes rather than across the full distribution. Trade/GDP is nearly uncorrelated with the financial cluster, consistent with trade openness being driven by geography and policy rather than domestic financial conditions.

Conclusions

The dataset covers 217 economies from 1960 to 2025. Its most consequential data quality issue is the non-random absence of financial variables—Market Cap/GDP is missing in 81.2% of country-years, concentrated in lower-income economies—a selection bias that must be made explicit before fitting any model that includes these regressors. Log-transformation is not optional for GDP per Capita or Inflation, both of which span multiple orders of magnitude; hyperinflationary observations (Table 2) additionally require winsorisation or case-by-case treatment. The most relevant structural pattern is the tight co-movement of financial depth with per-capita income ($\rho > 0.7$), forming a latent dimension recoverable by PCA or factor analysis, alongside a persistent widening of Credit/GDP across income groups since the 1990s. Researchers extending this work in the cross-country growth empirics tradition (Dollar & Kraay, 2002) should plan for two-way fixed effects, explicit missingness handling, and potential enrichment with institutional indicators (Acemoglu, Johnson & Robinson, 2001) to separate financial

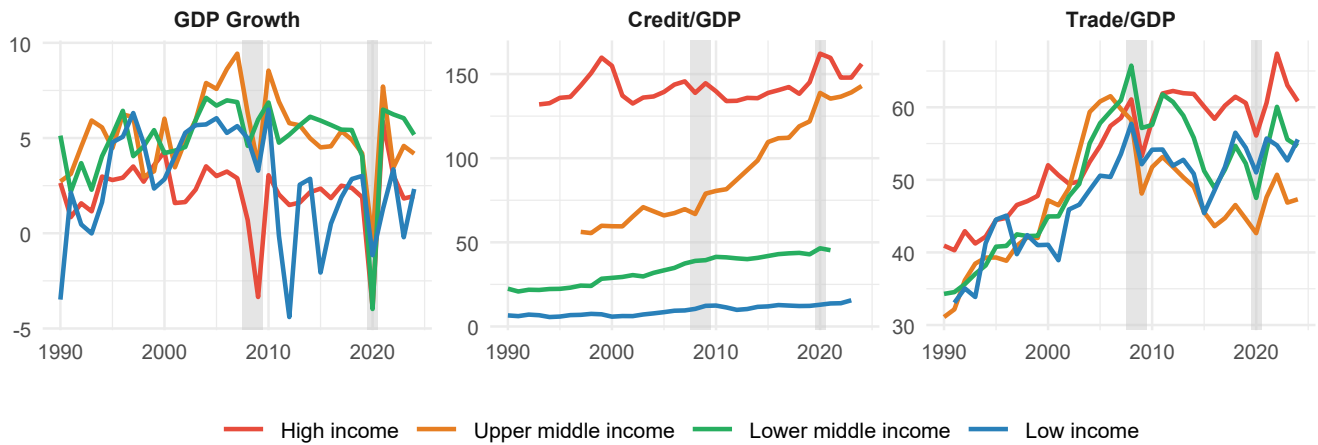


Figure 3: Trends (1990–2023) for three indicators across World Bank income groups. Grey bands mark the Global Financial Crisis (2008–2009) and the COVID-19 shock (2020). Income-group series are World Bank composite aggregates and do not require individual-country imputation.

development from governance effects.

References

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Figure 4: Spearman rank correlation matrix computed on pairwise-complete country-year observations. Spearman is used in preference to Pearson because GDP per Capita and Inflation are strongly right-skewed. Variables are reordered by hierarchical clustering. Blank cells indicate correlations not reaching significance after Bonferroni correction ($\alpha = 0.05/28$ pairs). Colour intensity reflects magnitude.